

Mesenchymal stem cell-derived extracellular vesicles: Recent therapeutics and targeted drug delivery advances

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Abstract

The targeted drug delivery field is rapidly advancing, focusing on developing biocompatible nanoparticles that meet rigorous criteria of non-toxicity, biocompatibility, and efficient release of encapsulated molecules. Conventional synthetic nanoparticles (SNPs) face complications such as elevated immune responses, complex synthesis methods, and toxicity, which restrict their utility in therapeutics and drug delivery. Extracellular vesicles (EVs) have emerged as promising substitutes for SNPs, leveraging their ability to cross biological barriers, biocompatibility, reduced toxicity, and natural origin. Notably, mesenchymal stem cell-derived EVs (MSC-EVs) have garnered much curiosity due to their potential in therapeutics and drug delivery. Studies suggest that MSC-EVs, the central paracrine contributors of MSCs, replicate the therapeutic effects of MSCs. This review explores the characteristics of MSC-EVs, emphasizing their potential in therapeutics and drug delivery for various diseases, including CRISPR/Cas9 delivery for gene editing. It also delves into the obstacles and challenges of MSC-EVs in clinical applications and provides insights into strategies to overcome the limitations of biodistribution and target delivery.